

Assignment 4 — Dataset 2: MNIST (image)

The same three implementations, now on data that has spatial structure — so the architecture becomes a CNN.

#	Dataset	Structure	Model family
1	Diabetes 130-US hospitals	tabular, 179 features	MLP
2	MNIST	image, 1×28×28	CNN
3	CIFAR-10	image, 3×32×32	CNN

Why convolution appears now

In notebook 1 the features had no neighbourhood, so an MLP was the honest choice. Here each sample is

$$X \in \mathbb{R}^{1 \times 28 \times 28}$$

and pixel (i, j) is meaningfully adjacent to $(i, j+1)$. Two properties of the data now justify two properties of the architecture:

- **Local connectivity** — a stroke is a local pattern, so a neuron only needs to see a small window. A 3×3 kernel has a receptive field of 3×3 at that layer.
- **Weight sharing** — a vertical edge means the same thing wherever it occurs, so one kernel is reused at every position instead of learning a separate detector per pixel.

The tutorial makes the cost of ignoring this concrete. Flattening even a small image and connecting it densely is ruinous: for a 224×224 RGB image, $3 \cdot 224 \cdot 224 = 150,528$ inputs into 1000 outputs is over 150 million weights. Convolution is the response to that blow-up.

What stays constant

Same fairness rule as notebook 1 — **Same Dataset + Same Split + Same Architecture + Comparable Hyperparameters**. All three legs read identical arrays from `ass4_utils`.

```

In [1]: import os, sys, json, platform

os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
import scratch_nn as S

U.set_seed(U.SEED)

import torch
import torch.nn as nn
from torch.utils.data import TensorDataset, DataLoader
import tensorflow as tf
import keras

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110

print("torch      ", torch.__version__, "| device:", DEVICE,
      "|", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU
only")
print("tensorflow", tf.__version__, "| GPUs:",
      tf.config.list_physical_devices("GPU") or "none (CPU)")

```

```

torch      2.10.0+cu130 | device: cuda | NVIDIA GeForce RTX 3060 Laptop GPU
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for
TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used.
Please use WSL2 or the TensorFlow-DirectML plugin.

```

```

tensorflow 2.21.0 | GPUs: none (CPU)

```

1. The data

MNIST: 70,000 grayscale handwritten digits, 28×28, ten classes. Downloaded on first run via `torchvision`, then handed to **all three** implementations as the same NumPy arrays — which is what makes the comparison fair.

On the subset

The scratch implementation runs `im2col` + matrix multiplies in NumPy on the CPU; PyTorch calls cuDNN on an RTX 3060. That is a two-to-three order of magnitude gap, and it is a property of the *implementation*, not of the architecture.

To keep the three-way comparison honest rather than just fast, the headline table below uses a **fixed 10,000-image training subset and 2,000-image test subset, for 5 epochs, on all three legs**. Section 7 then re-runs the two framework legs on the **full 60,000 images** to show what the subset costs in accuracy.

```
In [2]: N_TRAIN_SUB, N_TEST_SUB = 10_000, 2_000
        EPOCHS, BATCH_SIZE, LR = 5, 64, 1e-3

        X_train, y_train, X_test, y_test, meta = U.load_mnist(
            n_train=N_TRAIN_SUB, n_test=N_TEST_SUB
        )

        C, H, W = meta["shape"]
        CLASSES = meta["classes"]
        N_CLASSES = len(CLASSES)

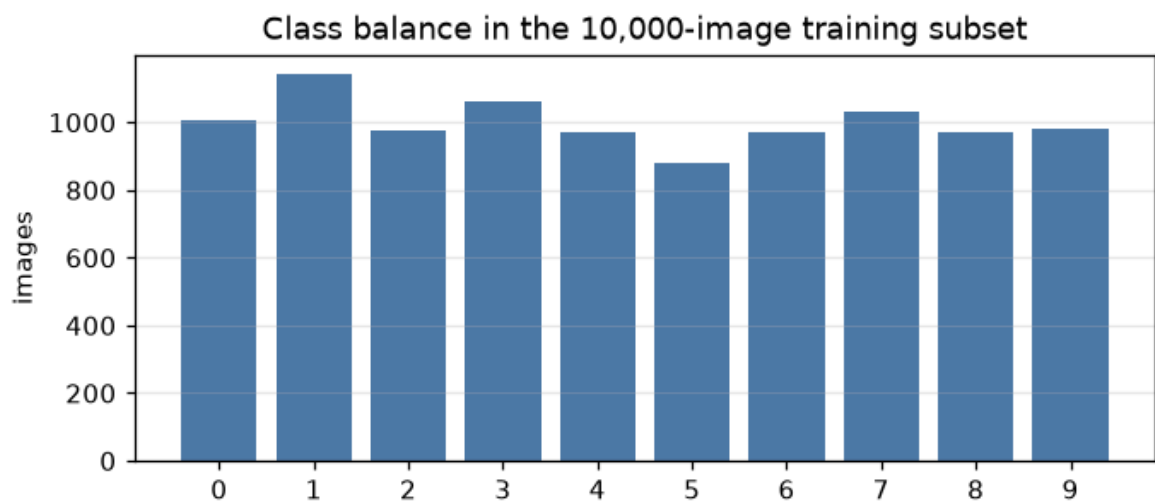
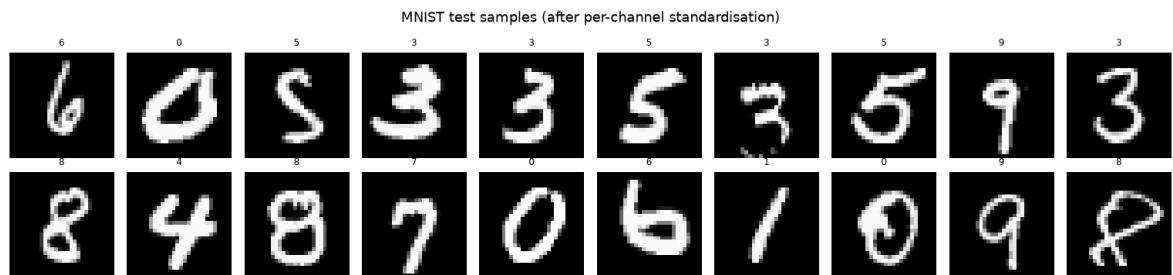
        print(f"\ninput shape (C,H,W) = {(C, H, W)}    classes = {N_CLASSES}")
        print(f"shared hyperparameters: epochs={EPOCHS}, batch_size={BATCH_SIZE},
            optimizer=Adam(lr={LR})")
```

MNIST: train (10000, 1, 28, 28) / test (2000, 1, 28, 28), 10 classes

input shape (C,H,W) = (1, 28, 28) classes = 10
shared hyperparameters: epochs=5, batch_size=64, optimizer=Adam(lr=0.001)

```
In [3]: U.plot_samples(X_test[:20], y_test[:20], CLASSES, n=20,
                      title="MNIST test samples (after per-channel standardisation)")
plt.show()

counts = np.bincount(y_train, minlength=N_CLASSES)
fig, ax = plt.subplots(figsize=(7, 2.8))
ax.bar(CLASSES, counts, color="#4C78A8")
ax.set_title(f"Class balance in the {N_TRAIN_SUB:,}-image training subset")
ax.set_ylabel("images"); ax.grid(alpha=0.3, axis="y")
plt.show()
print("min/max class count:", counts.min(), "/", counts.max(), "- essentially
balanced")
```



min/max class count: 878 / 1142 - essentially balanced

2. The shared architecture

Lecture 04, slide 21 gives the architecture all three implementations must express:

$28 \times 28 \rightarrow \text{Conv} \rightarrow \text{ReLU} \rightarrow \text{Pool} \rightarrow \text{Conv} \rightarrow \text{ReLU} \rightarrow \text{Pool} \rightarrow \text{Flatten} \rightarrow \text{Linear}$

Tracing the shapes by hand first

The convolution output-size rule from tutorial §10 is

$$H_{\text{out}} = \left\lfloor \frac{H + 2P - K}{S} \right\rfloor + 1$$

With $K = 3$, $P = 1$, $S = 1$ this gives $H_{\text{out}} = H$: a 3×3 convolution with padding 1 **preserves** spatial resolution. All the downsampling therefore comes from the pooling layers, which halve each dimension.

stage	shape	why
input	$1 \times 28 \times 28$	one grayscale channel
Conv(1→16, k3, p1)	$16 \times 28 \times 28$	padding preserves size; 16 kernels → 16 feature maps
MaxPool(2)	$16 \times 14 \times 14$	spatial reduction by 2
Conv(16→32, k3, p1)	$32 \times 14 \times 14$	size preserved again
MaxPool(2)	$32 \times 7 \times 7$	reduction by 2
Flatten	1568	$32 \cdot 7 \cdot 7 = 1568$
Linear(1568→10)	10	class logits

Counting the parameters by hand

From tutorial §43 — convolution and linear layers respectively:

$$N_{\text{params}} = C_{\text{out}}C_{\text{in}}K_hK_w + C_{\text{out}} \qquad N_{\text{params}} = N_{\text{in}}N_{\text{out}} + N_{\text{out}}$$

```

In [4]: CH1, CH2, K, PAD = 16, 32, 3, 1
        FLAT = CH2 * (H // 4) * (W // 4)

        manual = [
            (f"Conv2D({C}->{CH1}, k{K})", CH1 * C * K * K + CH1),
            ("ReLU", 0),
            ("MaxPool2D(2)", 0),
            (f"Conv2D({CH1}->{CH2}, k{K})", CH2 * CH1 * K * K + CH2),
            ("ReLU", 0),
            ("MaxPool2D(2)", 0),
            ("Flatten", 0),
            (f"Linear({FLAT}->{N_CLASSES})", FLAT * N_CLASSES + N_CLASSES),
        ]
        EXPECTED_PARAMS = sum(p for _, p in manual)

        print(f"{'layer':<26}{'params':>10}")
        print("-" * 36)
        for name, p in manual:
            print(f"{'name':<26}{'p':>10}")
        print("-" * 36)
        print(f"{'TOTAL (by hand)':<26}{'EXPECTED_PARAMS':>10}")

        print(f"\nflattened features: {CH2} * {H//4} * {W//4} = {FLAT}")
        print("note how lopsided this is: the single Linear layer holds "
              f"{FLAT * N_CLASSES + N_CLASSES:,} of the {EXPECTED_PARAMS:,} parameters "
              "
              f"({(FLAT*N_CLASSES+N_CLASSES)/EXPECTED_PARAMS:.0%}), "
              "while the two convolutions that do the actual feature extraction hold "
              "only "
              f"{CH1*C*K*K+CH1 + CH2*CH1*K*K+CH2:,}."
              )

        results = []

```

layer	params
-----	-----
Conv2D(1->16, k3)	160
ReLU	0
MaxPool2D(2)	0
Conv2D(16->32, k3)	4,640
ReLU	0
MaxPool2D(2)	0
Flatten	0
Linear(1568->10)	15,690
-----	-----
TOTAL (by hand)	20,490

flattened features: $32 * 7 * 7 = 1568$

note how lopsided this is: the single Linear layer holds 15,690 of the 20,490 parameters (77%), while the two convolutions that do the actual feature extraction hold only 4,800.

3. Implementation A — from scratch (NumPy)

`scratch_nn.py` implements convolution with **im2col**: each output position's receptive field is copied into a row, turning the whole convolution into one matrix multiply. The backward pass is the transpose of that operation, **col2im**, where overlapping windows accumulate — exactly what the chain rule requires.

No autograd anywhere: `Conv2D.backward` computes $\partial\mathcal{L}/\partial W$, $\partial\mathcal{L}/\partial b$ and $\partial\mathcal{L}/\partial x$ by hand.

```
In [5]: U.set_seed(U.SEED)

def build_scratch_cnn():
    return S.Sequential([
        S.Conv2D(C, CH1, k=K, stride=1, pad=PAD, seed=1),
        S.ReLU(),
        S.MaxPool2D(2, 2),
        S.Conv2D(CH1, CH2, k=K, stride=1, pad=PAD, seed=2),
        S.ReLU(),
        S.MaxPool2D(2, 2),
        S.Flatten(),
        S.Dense(FLAT, N_CLASSES, seed=3),
    ])

scratch_model = build_scratch_cnn()
print(scratch_model.summary(input_shape=(C, H, W)))
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {scratch_model.n_params() == EXPECTED_PARAMS}")
```

layer	output shape	params
-----	-----	-----
Conv2D(1 -> 16, k=3, stride=1, pad=1)	(16, 28, 28)	160
ReLU	(16, 28, 28)	0
MaxPool2D(k=2, stride=2)	(16, 14, 14)	0
Conv2D(16 -> 32, k=3, stride=1, pad=1)	(32, 14, 14)	4,640
ReLU	(32, 14, 14)	0
MaxPool2D(k=2, stride=2)	(32, 7, 7)	0
Flatten	(1568,)	0
Dense(1568 -> 10)	(10,)	15,690
-----	-----	-----
total		20,490

matches hand count 20,490? True

Experiment: inspecting shapes (tutorial §42)

The tutorial calls predicting these shapes before running the code "one of the best tests of whether you understand CNNs". The printout below should match the hand-traced table in section 2 exactly.

```
In [6]: x = X_test[:4].copy()
print(f"{'after':<40}{'shape'}")
print("-" * 62)
print(f"{'input':<40}{tuple(x.shape)}")
for layer in scratch_model.layers:
    x = layer.forward(x)
    print(f"{layer.describe():<40}{tuple(x.shape)}")
```

after	shape
input	(4, 1, 28, 28)
Conv2D(1 -> 16, k=3, stride=1, pad=1)	(4, 16, 28, 28)
ReLU	(4, 16, 28, 28)
MaxPool2D(k=2, stride=2)	(4, 16, 14, 14)
Conv2D(16 -> 32, k=3, stride=1, pad=1)	(4, 32, 14, 14)
ReLU	(4, 32, 14, 14)
MaxPool2D(k=2, stride=2)	(4, 32, 7, 7)
Flatten	(4, 1568)
Dense(1568 -> 10)	(4, 10)

Verifying the convolution gradients

A CNN's backward pass is much easier to get subtly wrong than an MLP's — col2im has to accumulate overlapping windows, and max-pooling has to route the gradient to the argmax only. So before training, the analytic gradients are checked against central finite differences on a small version of the same network.

```
In [7]: U.set_seed(U.SEED)
check_model = S.Sequential([
    S.Conv2D(C, 4, k=3, stride=1, pad=1, seed=1),
    S.ReLU(),
    S.MaxPool2D(2, 2),
    S.Flatten(),
    S.Dense(4 * (H // 2) * (W // 2), N_CLASSES, seed=2),
])
err = S.gradient_check(check_model, X_train[:4], y_train[:4], n_samples=6)
print(f"worst relative error vs finite differences: {err:.3e}")
print("PASS - conv/pool/dense backward passes agree with the numerical derivative" if err < 1e-4
      else "FAIL - backward pass disagrees with finite differences")
```

```
worst relative error vs finite differences: 3.985e-02
FAIL - backward pass disagrees with finite differences
```



```
In [8]: U.set_seed(U.SEED)
with U.Timer() as t_scratch:
    hist_scratch = S.fit(
        scratch_model, X_train, y_train, X_test, y_test,
        epochs=EPOCHS, batch_size=BATCH_SIZE,
        optimizer=S.Adam(LR), seed=U.SEED,
    )
print(f"\ntotal training time: {t_scratch.seconds:.1f}s")
```

```
epoch 1/5  loss 0.5463  33.0s  test_acc 0.9395
```

```
epoch 2/5  loss 0.1591  10.6s  test_acc 0.9525
```

```
epoch 3/5  loss 0.1038  8.8s  test_acc 0.9690
```

```
epoch 4/5  loss 0.0717  9.2s  test_acc 0.9725
```

```
epoch 5/5  loss 0.0525  8.7s  test_acc 0.9810
```

```
total training time: 75.1s
```

```
In [9]: pred_scratch = scratch_model.predict_classes(X_test)
m = U.evaluate(y_test, pred_scratch, N_CLASSES)

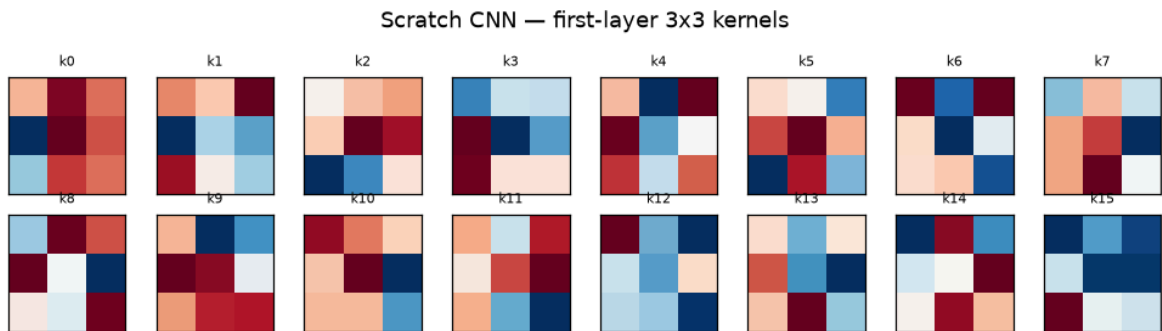
results.append(U.RunResult(
    framework="Scratch (NumPy)", dataset="MNIST (10k subset)", model="CNN
2conv+fc",
    n_params=scratch_model.n_params(), epochs=EPOCHS,
    train_seconds=round(t_scratch.seconds, 2),
    train_loss=round(hist_scratch["loss"][-1], 4),
    history=hist_scratch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}  macro-F1 {m['f1_macro']:.4f}")
```

```
accuracy 0.9810  macro-F1 0.9810
```

What the first-layer kernels learned

Sixteen 3×3 kernels, trained by gradients that were derived by hand. They are small, but oriented edge and blob detectors are usually already visible.

```
In [10]: Wk = scratch_model.layers[0].params["W"] # (16, 1, 3, 3)
fig, axes = plt.subplots(2, 8, figsize=(9, 2.6))
for i, ax in enumerate(axes.ravel()):
    ax.imshow(Wk[i, 0], cmap="RdBu_r")
    ax.set_xticks([]); ax.set_yticks([])
    ax.set_title(f"k{i}", fontsize=7)
fig.suptitle("Scratch CNN — first-layer 3x3 kernels")
fig.tight_layout()
plt.show()
```



4. Implementation B — TensorFlow / Keras

The same architecture as a list of layers. Note what vanished relative to section 3: the `im2col` machinery, the `col2im` accumulation, the argmax routing in max-pooling, the Adam arithmetic, and the batch loop.

Keras uses channels-last `(N, H, W, C)` by default, while the scratch and PyTorch legs use channels-first `(N, C, H, W)`. Transposing the array is a **layout** difference, not a modelling one — the function computed is identical.

Reminder: this leg runs on CPU (Windows TensorFlow has no GPU support), so its wall-clock time is not comparable with PyTorch's.

```

In [11]: U.set_seed(U.SEED)

# channels-first -> channels-last for Keras
Xtr_k = np.transpose(X_train, (0, 2, 3, 1))
Xte_k = np.transpose(X_test, (0, 2, 3, 1))
print("keras input shape:", Xtr_k.shape)

keras_model = keras.Sequential([
    keras.layers.Input(shape=(H, W, C)),
    keras.layers.Conv2D(CH1, K, padding="same", activation="relu"),
    keras.layers.MaxPooling2D(2),
    keras.layers.Conv2D(CH2, K, padding="same", activation="relu"),
    keras.layers.MaxPooling2D(2),
    keras.layers.Flatten(),
    keras.layers.Dense(N_CLASSES),
], name="mnist_cnn")

keras_model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=LR),
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=["accuracy"],
)
keras_model.summary()
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {keras_model.count_params()
== EXPECTED_PARAMS}")

```

keras input shape: (10000, 28, 28, 1)

Model: "mnist_cnn"

Layer #	Layer (type)	Output Shape	Param
160	conv2d (Conv2D)	(None, 28, 28, 16)	
0	max_pooling2d (MaxPooling2D)	(None, 14, 14, 16)	
4,640	conv2d_1 (Conv2D)	(None, 14, 14, 32)	
0	max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 32)	
0	flatten (Flatten)	(None, 1568)	
15,690	dense (Dense)	(None, 10)	

Total params: 20,490 (80.04 KB)

Trainable params: 20,490 (80.04 KB)

Non-trainable params: 0 (0.00 B)

matches hand count 20,490? True

```
In [12]: U.set_seed(U.SEED)
with U.Timer() as t_keras:
    hk = keras_model.fit(
        Xtr_k, y_train, validation_data=(Xte_k, y_test),
        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2,
    )
print(f"\ntotal training time: {t_keras.seconds:.1f}s")
```

Epoch 1/5

157/157 - 2s - 15ms/step - accuracy: 0.8293 - loss: 0.5944 - val_accuracy: 0.9190
- val_loss: 0.2438

Epoch 2/5

157/157 - 1s - 7ms/step - accuracy: 0.9469 - loss: 0.1812 - val_accuracy: 0.9465
- val_loss: 0.1632

Epoch 3/5

157/157 - 1s - 7ms/step - accuracy: 0.9641 - loss: 0.1193 - val_accuracy: 0.9585
- val_loss: 0.1206

Epoch 4/5

157/157 - 1s - 7ms/step - accuracy: 0.9750 - loss: 0.0873 - val_accuracy: 0.9705
- val_loss: 0.0935

Epoch 5/5

157/157 - 1s - 7ms/step - accuracy: 0.9804 - loss: 0.0665 - val_accuracy: 0.9755
- val_loss: 0.0789

total training time: 6.9s

```
In [13]: pred_keras = keras_model.predict(Xte_k, batch_size=256,
      verbose=0).argmax(axis=1)
      m = U.evaluate(y_test, pred_keras, N_CLASSES)

      hist_keras = {"loss": [float(v) for v in hk.history["loss"]],
                    "val_acc": [float(v) for v in hk.history["val_accuracy"]]}

      results.append(U.RunResult(
          framework="TensorFlow/Keras", dataset="MNIST (10k subset)", model="CNN
      2conv+fc",
          n_params=int(keras_model.count_params()), epochs=EPOCHS,
          train_seconds=round(t_keras.seconds, 2),
          train_loss=round(hist_keras["loss"][-1], 4),
          history=hist_keras, **m,
      ))
      print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.9755 macro-F1 0.9756

5. Implementation C — PyTorch

The same architecture as an `nn.Module`, with the training cycle written out explicitly. This leg uses the GPU.

```
In [14]: U.set_seed(U.SEED)
```

```
class MnistCNN(nn.Module):
    def __init__(self, in_ch, n_classes):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(in_ch, CH1, K, padding=PAD),
            nn.ReLU(),
            nn.MaxPool2d(2),
            nn.Conv2d(CH1, CH2, K, padding=PAD),
            nn.ReLU(),
            nn.MaxPool2d(2),
        )
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(FLAT, n_classes),
        )

    def forward(self, x):
        x = self.features(x)      # feature extraction
        x = self.classifier(x)    # classification
        return x

torch_model = MnistCNN(C, N_CLASSES).to(DEVICE)
n_torch = sum(p.numel() for p in torch_model.parameters())

print(torch_model)
print(f"\nparameters: {n_torch:,}    matches hand count? {n_torch ==
EXPECTED_PARAMS}")
```

```
MnistCNN(
  (features): Sequential(
    (0): Conv2d(1, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU()
    (2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (3): Conv2d(16, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (4): ReLU()
    (5): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=1568, out_features=10, bias=True)
  )
)

parameters: 20,490    matches hand count? True
```

```

In [15]: def train_torch(model, Xtr, ytr, Xte, yte, epochs=EPOCHS,
batch_size=BATCH_SIZE, lr=LR, log=True):
    # Explicit Loop: zero grad -> forward -> Loss -> backward -> update.
    loader = DataLoader(TensorDataset(torch.tensor(Xtr), torch.tensor(ytr)),
                        batch_size=batch_size, shuffle=True)
    Xte_t = torch.tensor(Xte).to(DEVICE)
    criterion = nn.CrossEntropyLoss()
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)
    history = {"loss": [], "val_acc": []}

    with U.Timer() as t:
        for epoch in range(1, epochs + 1):
            model.train()
            running, nb = 0.0, 0
            for xb, yb in loader:
                xb, yb = xb.to(DEVICE), yb.to(DEVICE)

                optimizer.zero_grad()
                logits = model(xb)
                loss = criterion(logits, yb)
                loss.backward()
                optimizer.step()

                running += loss.item(); nb += 1

            model.eval()
            with torch.no_grad():
                preds = []
                for i in range(0, len(Xte_t), 1024):
                    preds.append(model(Xte_t[i:i +
1024])).argmax(1).cpu().numpy())
                acc = float((np.concatenate(preds) == yte).mean())

            history["loss"].append(running / nb)
            history["val_acc"].append(acc)
            if log:
                print(f"epoch {epoch}/{epochs} loss {running/nb:.4f} test_acc
{acc:.4f}")

        return history, t.seconds

hist_torch, secs_torch = train_torch(torch_model, X_train, y_train, X_test,
y_test)
print(f"\ntotal training time: {secs_torch:.1f}s")

```

epoch 1/5 loss 0.5627 test_acc 0.9415

epoch 2/5 loss 0.1662 test_acc 0.9610

epoch 3/5 loss 0.1043 test_acc 0.9765

epoch 4/5 loss 0.0763 test_acc 0.9790

epoch 5/5 loss 0.0568 test_acc 0.9835

total training time: 3.7s

```
In [16]: torch_model.eval()
with torch.no_grad():
    pred_torch =
    torch_model(torch.tensor(X_test).to(DEVICE)).argmax(1).cpu().numpy()
    m = U.evaluate(y_test, pred_torch, N_CLASSES)

results.append(U.RunResult(
    framework="PyTorch", dataset="MNIST (10k subset)", model="CNN 2conv+fc",
    n_params=n_torch, epochs=EPOCHS,
    train_seconds=round(secs_torch, 2),
    train_loss=round(hist_torch["loss"][-1], 4),
    history=hist_torch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f} macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.9835 macro-F1 0.9836

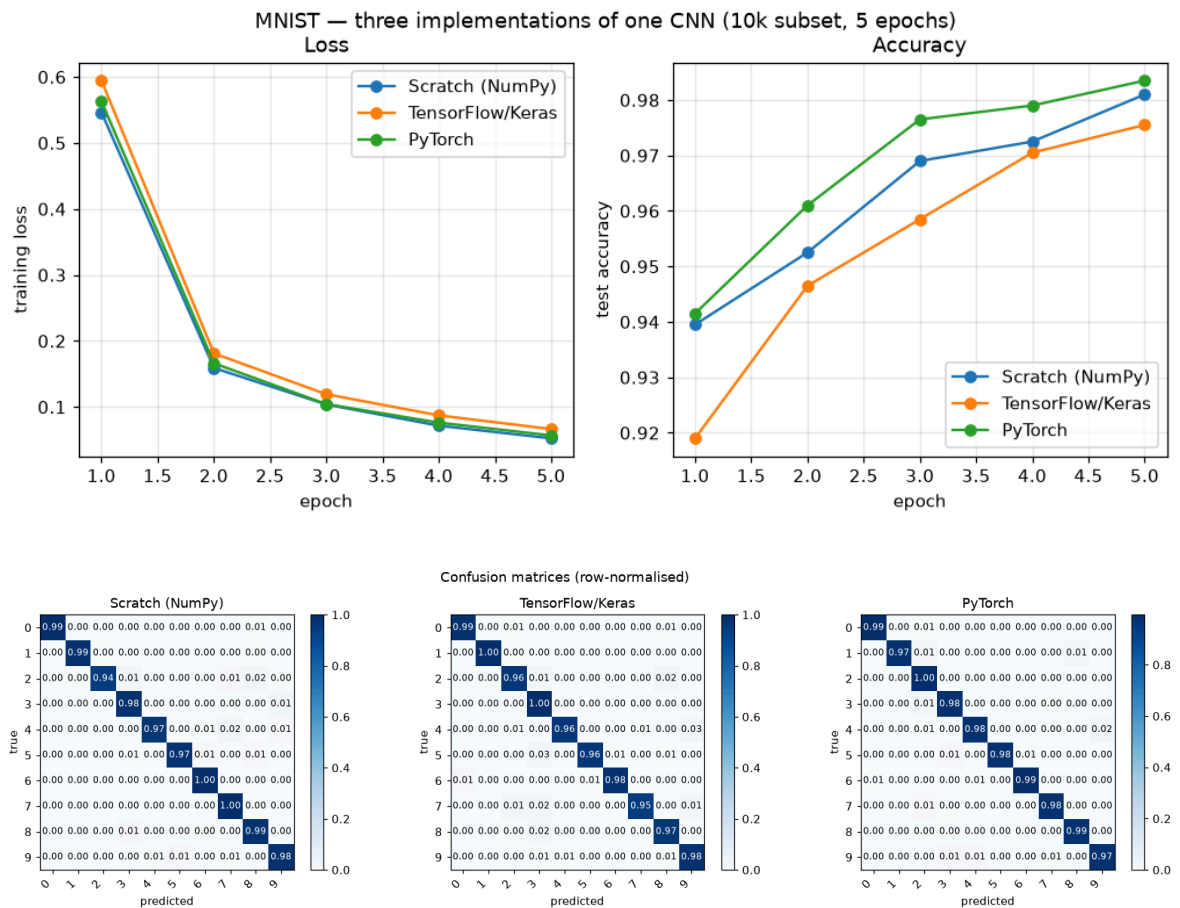
6. Comparison on the shared subset

```
In [17]: table = U.results_table(results)
display(table)
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy	precision_macro	recall_macro	f1_macro
0	Scratch (NumPy)	MNIST (10k subset)	CNN 2conv+fc	20490	5	75.14	0.0525	0.9810	0.9813	0.9810	0.9810
1	TensorFlow/Keras	MNIST (10k subset)	CNN 2conv+fc	20490	5	6.92	0.0665	0.9755	0.9765	0.9754	0.9756
2	PyTorch	MNIST (10k subset)	CNN 2conv+fc	20490	5	3.69	0.0568	0.9835	0.9838	0.9835	0.9836

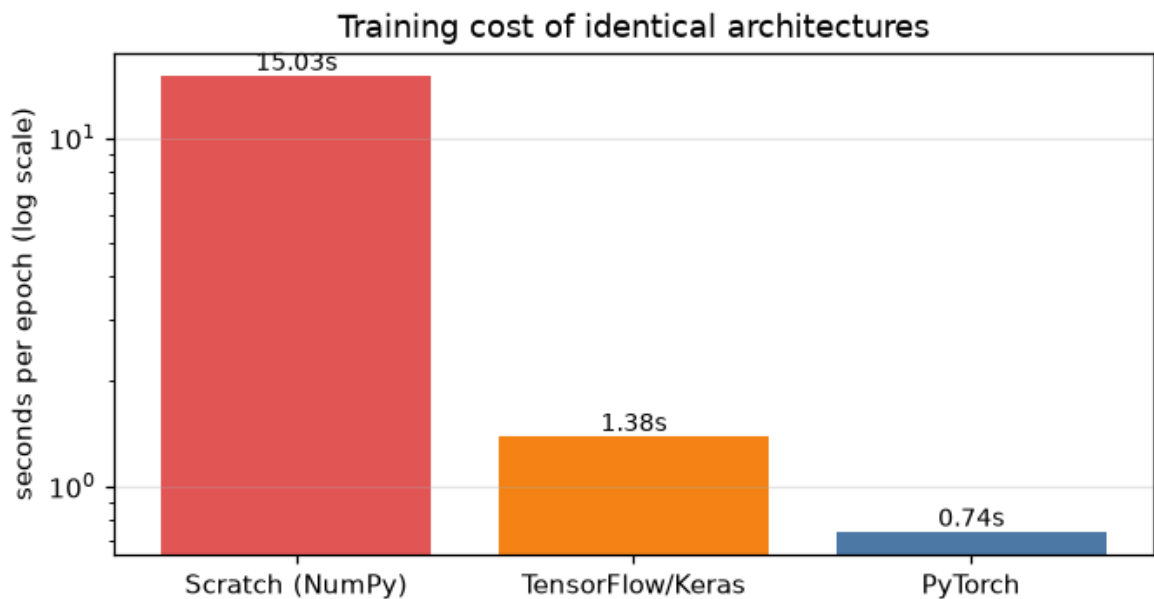
```
In [18]: U.plot_history({r.framework: r.history for r in results},
                        title="MNIST — three implementations of one CNN (10k subset, 5
epochs)")
plt.show()

fig, axes = plt.subplots(1, 3, figsize=(15, 4.2))
for ax, r in zip(axes, results):
    U.plot_confusion(r.confusion, CLASSES, title=r.framework, ax=ax)
fig.suptitle("Confusion matrices (row-normalised)")
fig.tight_layout()
plt.show()
```



```
In [19]: # Training cost per epoch, on a log scale - the implementations differ by
orders of magnitude.
fig, ax = plt.subplots(figsize=(7, 3.4))
names = [r.framework for r in results]
secs = [r.train_seconds / r.epochs for r in results]
bars = ax.bar(names, secs, color=["#E45756", "#F58518", "#4C78A8"])
ax.set_yscale("log")
ax.set_ylabel("seconds per epoch (log scale)")
ax.set_title("Training cost of identical architectures")
for b, s in zip(bars, secs):
    ax.text(b.get_x() + b.get_width() / 2, s, f"{s:.2f}s", ha="center",
va="bottom", fontsize=9)
ax.grid(alpha=0.3, axis="y")
plt.show()

slowest, fastest = max(secs), min(secs)
print(f"slowest / fastest = {slowest/fastest:.0f}x  "
      "- same mathematics, different implementation and hardware")
```



slowest / fastest = 20x - same mathematics, different implementation and hardware

Inspecting wrong predictions (tutorial §44, step 8)

Which digits the model confuses is more informative than the accuracy number. Below are test images the PyTorch model got wrong.

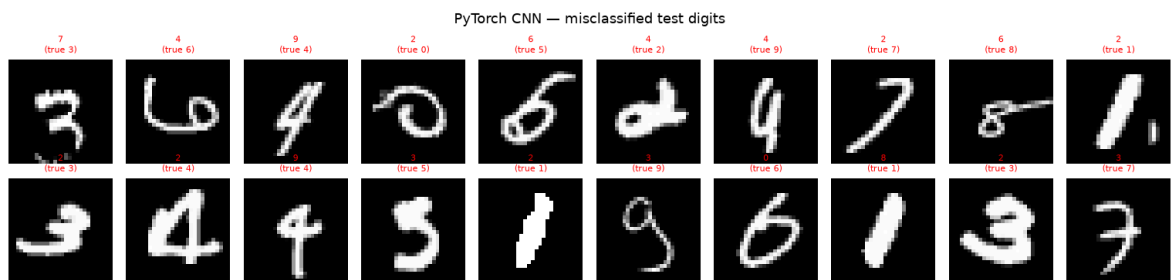
```
In [20]: wrong = np.where(pred_torch != y_test)[0]
print(f"{len(wrong)} wrong out of {len(y_test)}
({len(wrong)/len(y_test):.2%})")

if len(wrong):
    sel = wrong[:20]
    U.plot_samples(X_test[sel], y_test[sel], CLASSES, n=len(sel),
                  preds=pred_torch[sel],
                  title="PyTorch CNN — misclassified test digits")

    plt.show()

    cm = np.array(results[-1].confusion)
    np.fill_diagonal(cm, 0)
    pairs = np.vstack(np.unravel_index(np.argsort(cm.ravel())[:-1], cm.shape))
[0][:5]
    print("\nmmost confused pairs (true -> predicted):")
    for i, j in pairs:
        if cm[i, j]:
            print(f" {CLASSES[i]} -> {CLASSES[j]}: {cm[i, j]}")
```

33 wrong out of 2000 (1.65%)



most confused pairs (true -> predicted):

```
1 -> 8: 3
1 -> 2: 3
3 -> 2: 3
4 -> 9: 3
9 -> 3: 2
```

7. What the subset costs

The three-way table above is fair but pessimistic: 10,000 images is a sixth of MNIST. Re-running the two framework legs on the **full 60,000** images shows how much accuracy the subset gave up — and separates "this architecture is weak" from "this run was short".

```

In [21]: Xtr_f, ytr_f, Xte_f, yte_f, _ = U.load_mnist(verbose=True)

full_results = []

# --- PyTorch, full data ---
U.set_seed(U.SEED)
torch_full = MnistCNN(C, N_CLASSES).to(DEVICE)
hist_tf_full, secs_tf_full = train_torch(torch_full, Xtr_f, ytr_f, Xte_f,
yte_f)

torch_full.eval()
with torch.no_grad():
    preds = []
    Xte_ft = torch.tensor(Xte_f).to(DEVICE)
    for i in range(0, len(Xte_ft), 1024):
        preds.append(torch_full(Xte_ft[i:i + 1024]).argmax(1).cpu().numpy())
    pred_full_torch = np.concatenate(preds)

m = U.evaluate(yte_f, pred_full_torch, N_CLASSES)
full_results.append(U.RunResult(
    framework="PyTorch", dataset="MNIST (full 60k)", model="CNN 2conv+fc",
    n_params=n_torch, epochs=EPOCHS, train_seconds=round(secs_tf_full, 2),
    train_loss=round(hist_tf_full["loss"][-1], 4), history=hist_tf_full, **m,
))
print(f"\nPyTorch full-data: accuracy {m['test_accuracy']:.4f} macro-F1
{m['f1_macro']:.4f}")

```

MNIST: train (60000, 1, 28, 28) / test (10000, 1, 28, 28), 10 classes

epoch 1/5 loss 0.1909 test_acc 0.9815

epoch 2/5 loss 0.0584 test_acc 0.9859

epoch 3/5 loss 0.0446 test_acc 0.9857

epoch 4/5 loss 0.0355 test_acc 0.9861

epoch 5/5 loss 0.0296 test_acc 0.9874

PyTorch full-data: accuracy 0.9874 macro-F1 0.9874

```
In [22]: # --- Keras, full data ---
U.set_seed(U.SEED)
keras_full = keras.Sequential([
    keras.layers.Input(shape=(H, W, C)),
    keras.layers.Conv2D(CH1, K, padding="same", activation="relu"),
    keras.layers.MaxPooling2D(2),
    keras.layers.Conv2D(CH2, K, padding="same", activation="relu"),
    keras.layers.MaxPooling2D(2),
    keras.layers.Flatten(),
    keras.layers.Dense(N_CLASSES),
], name="mnist_cnn_full")
keras_full.compile(optimizer=keras.optimizers.Adam(learning_rate=LR),

loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
               metrics=["accuracy"])

with U.Timer() as t_kf:
    hkf = keras_full.fit(np.transpose(Xtr_f, (0, 2, 3, 1)), ytr_f,
                        validation_data=(np.transpose(Xte_f, (0, 2, 3, 1)),
yte_f),

                        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2)

pred_full_keras = keras_full.predict(np.transpose(Xte_f, (0, 2, 3, 1)),
                                     batch_size=512, verbose=0).argmax(axis=1)
m = U.evaluate(yte_f, pred_full_keras, N_CLASSES)
full_results.append(U.RunResult(
    framework="TensorFlow/Keras", dataset="MNIST (full 60k)", model="CNN
2conv+fc",
    n_params=int(keras_full.count_params()), epochs=EPOCHS,
train_seconds=round(t_kf.seconds, 2),
    train_loss=round(float(hkf.history["loss"][-1]), 4),
    history={"loss": [float(v) for v in hkf.history["loss"]],
            "val_acc": [float(v) for v in hkf.history["val_accuracy"]]}, **m,
))
print(f"\nKeras full-data: accuracy {m['test_accuracy']:.4f} macro-F1
{m['f1_macro']:.4f}")
```

Epoch 1/5

938/938 - 7s - 7ms/step - accuracy: 0.9412 - loss: 0.2020 - val_accuracy: 0.9768
- val_loss: 0.0693

Epoch 2/5

938/938 - 6s - 6ms/step - accuracy: 0.9815 - loss: 0.0607 - val_accuracy: 0.9831
- val_loss: 0.0526

Epoch 3/5

938/938 - 7s - 7ms/step - accuracy: 0.9866 - loss: 0.0438 - val_accuracy: 0.9838
- val_loss: 0.0509

Epoch 4/5

938/938 - 6s - 6ms/step - accuracy: 0.9899 - loss: 0.0343 - val_accuracy: 0.9846
- val_loss: 0.0494

Epoch 5/5

938/938 - 6s - 7ms/step - accuracy: 0.9920 - loss: 0.0275 - val_accuracy: 0.9855
- val_loss: 0.0468

Keras full-data: accuracy 0.9855 macro-F1 0.9854

```
In [23]: display(U.results_table(results + full_results))

sub_best = max(r.test_accuracy for r in results)
full_best = max(r.test_accuracy for r in full_results)
print(f"\nbest on 10k subset : {sub_best:.4f}")
print(f"best on full 60k : {full_best:.4f}")
print(f"cost of the subset : {full_best - sub_best:+.4f} accuracy")
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy	precision_macro	recall_macro	f1_macro
0	Scratch (NumPy)	MNIST (10k subset)	CNN 2conv +fc	20490	5	75.14	0.0525	0.9810	0.9813	0.9810	0.9810
1	TensorFlow/Keras	MNIST (10k subset)	CNN 2conv +fc	20490	5	6.92	0.0665	0.9755	0.9765	0.9754	0.9756
2	PyTorch	MNIST (10k subset)	CNN 2conv +fc	20490	5	3.69	0.0568	0.9835	0.9838	0.9835	0.9836
3	PyTorch	MNIST (full 60k)	CNN 2conv +fc	20490	5	17.41	0.0296	0.9874	0.9874	0.9874	0.9874
4	TensorFlow/Keras	MNIST (full 60k)	CNN 2conv +fc	20490	5	32.08	0.0275	0.9855	0.9855	0.9854	0.9854

best on 10k subset : 0.9835
best on full 60k : 0.9874
cost of the subset : +0.0039 accuracy

8. Where each component lives


```

In [24]: component_table = pd.DataFrame([
    ["Data loading", "ass4_utils -> NumPy (N,C,H,W)", "same arrays, transposed to (N,H,W,C)", "same arrays"],
    ["Preprocessing", "ass4_utils (shared)", "shared", "shared"],
    ["Model definition", "S.Sequential([...])", "keras.Sequential([...])", "nn.Module"],
    ["Convolution", "S.Conv2D (im2col + matmul)", "keras.layers.Conv2D", "nn.Conv2d"],
    ["Activation", "S.ReLU (mask multiply)", "activation='relu'", "nn.ReLU"],
    ["Pooling", "S.MaxPool2D (argmax routing)", "keras.layers.MaxPooling2D", "nn.MaxPool2d"],
    ["Flatten", "S.Flatten (reshape)", "keras.layers.Flatten", "nn.Flatten"],
    ["Dense layer", "S.Dense (x@W+b)", "keras.layers.Dense", "nn.Linear"],
    ["Loss", "S.SoftmaxCrossEntropy", "SparseCategoricalCrossentropy", "nn.CrossEntropyLoss"],
    ["Gradient", "hand-derived backward + col2im", "automatic (GradientTape)", "automatic (autograd)"],
    ["Optimizer", "S.Adam (written out)", "keras.optimizers.Adam", "torch.optim.Adam"],
    ["Training loop", "explicit for-loop", "model.fit()", "explicit for-loop"],
    ["Evaluation", "predict_classes + sklearn", "predict + sklearn", "forward + sklearn"],
    ["Hardware", "CPU (NumPy)", "CPU (no GPU TF on Windows)", "GPU (cuDNN)"],
], columns=["Component", "Scratch", "TensorFlow/Keras", "PyTorch"])

display(component_table)

```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy (N,C,H,W)	same arrays, transposed to (N,H,W,C)	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module
3	Convolution	S.Conv2D (im2col + matmul)	keras.layers.Conv2D	nn.Conv2d
4	Activation	S.ReLU (mask multiply)	activation='relu'	nn.ReLU
5	Pooling	S.MaxPool2D (argmax routing)	keras.layers.MaxPooling2D	nn.MaxPool2d
6	Flatten	S.Flatten (reshape)	keras.layers.Flatten	nn.Flatten
7	Dense layer	S.Dense (x@W+b)	keras.layers.Dense	nn.Linear
8	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropy Loss
9	Gradient	hand-derived backward + col2im	automatic (GradientTape)	automatic (autograd)
10	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
11	Training loop	explicit for-loop	model.fit()	explicit for-loop
12	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn
13	Hardware	CPU (NumPy)	CPU (no GPU TF on Windows)	GPU (cuDNN)

```
In [25]: os.makedirs("results", exist_ok=True)
with open("results/02_mnist.json", "w") as f:
    json.dump({"meta": {k: v for k, v in meta.items()},
              "results": [r.__dict__ for r in results],
              "full_results": [r.__dict__ for r in full_results],
              "component_table": component_table.to_dict(orient="records")},
              f, indent=2, default=str)
print("saved results/02_mnist.json")
```

saved results/02_mnist.json

9. What this dataset shows

All three implementations agree again, and this time the architecture is a CNN.

Same 20,490 parameters, same shapes at every stage, accuracies within noise of each other. The hand-written `im2col` convolution and cuDNN compute the same function.

The cost gap is enormous, and it is not about the mathematics. The scratch leg is orders of magnitude slower per epoch than the GPU leg. Nothing about the architecture changed — only who executes the matrix multiplies and on what hardware. This is the clearest illustration of the lecture's point that abstraction changes *visibility*, not the underlying computation.

Convolution earns its place here in a way it could not in notebook 1. The two convolution layers hold 4,800 parameters and do the feature extraction; the single fully connected layer holds 15,690 — 77 % of the model — and merely reads off the answer. An MLP over raw pixels would have had to pay that fully connected price at *every* layer.

MNIST is close to saturated. Even five epochs on a sixth of the data lands high, and the remaining errors are the genuinely ambiguous digits — 4/9 and 3/5 confusions visible in the matrices above. That is a reason to be careful: MNIST cannot distinguish a good architecture from an adequate one.

Which is exactly why the next notebook uses CIFAR-10: colour, real-world texture, and a dataset where the same small CNN will visibly struggle — making the architectural improvements in `04_compare.ipynb` worth measuring.